

A Reproducible Synthetic-Pigment Mixer for RGB Painting in Rust

pigment-mix project — independent implementation study

29 September 2026

Abstract

Ordinary color interpolation does not model the wavelength-dependent absorption and scattering of pigment mixtures. We investigate an RGB-input, RGB-output painting model built independently from public color-science equations. The implementation combines an eight-component synthetic pigment palette, constrained color-to-concentration inversion, and a linear-light reconstruction residual. The concentration-plus-residual principle is established prior work; our contribution is an inspectable implementation and reproducible assessment of its numerical and practical compromises. An 81-band double-precision reference is compared with a 21-band single-precision decoder and tetrahedral RGB lookup tables. Across 10000 seeded colors, the fast corrected reconstruction has mean CIEDE2000 1.19e-07 and maximum 2.38e-06; without the residual, the reference mean error is 11.7. The corrected endpoint accuracy follows from an algebraic identity and is not evidence of real-paint accuracy. Across 10000 random mixtures, the default 33³ fast model differs from direct reference inversion by mean 0.409 and 95th-percentile 1.19 CIEDE2000. Median native throughput is 1.66 million complete RGB mixtures per second, or 4.42 million with cached latents, on the recorded host. Canonical yellow–blue and red–blue trials produce green and dark purple, respectively. The worst default-LUT discrepancy is 40.1 CIEDE2000; significant outliers, palette-gamut limitations and steep trajectories remain. The release is a practical research basis for painting integration, not a calibrated simulator of specific commercial paints.

1. Introduction

A digital brush can perform millions of color combinations while a user expects the result to resemble a familiar material. Linear interpolation of encoded RGB is inexpensive, but its numerical coordinates combine a nonlinear display encoding rather than pigment amounts. Decoding the transfer function first makes interpolation appropriate for additive light, yet still does not model the repeated absorption and scattering that occur inside paint. Perceptual interpolation offers a different benefit: relatively regular changes in perceived lightness and chroma. It also lacks a model of a mixture’s material composition.

The difficulty is deeper than selecting a different interpo-

lation space. A color picked from an RGB image does not uniquely specify a reflectance spectrum. Even if that spectrum were known, opaque reflectance alone would determine a ratio of absorption to scattering, not their independent strengths. Two paints with identical isolated appearance can consequently behave differently when mixed with white or another pigment. A general RGB mixer must choose a material interpretation. The resulting model can be physically motivated without being a uniquely correct reconstruction of the input paint.

We implement a deliberately explicit interpretation. Smooth analytic reflectances define a small synthetic palette. Relative scattering strengths define tinting behavior. A deterministic constrained optimizer selects concentrations, and a residual restores the source RGB. This separates source compatibility from the nonlinear spectral transformation that bends mixture trajectories. Every synthetic parameter and generated table is available in the accompanying repository.

The project has three demonstrated contributions. First, it provides a dependency-free Rust runtime with separate reference and fast implementations of a documented synthetic-pigment model. Second, it provides deterministic model and LUT generation, numerical tests, raw measurements and figure scripts. Third, it evaluates endpoint consistency, uncorrected palette error, interpolation approximation, spectral sampling, runtime cost, and selected model ablations. We do not claim invention of the latent-residual method, optimal pigment selection, numerical compatibility with Mixbox, or physical accuracy against measured paint mixtures. These exclusions determine how the experiments should be read.

The scope is color mixing. The library does not simulate brush mechanics, layer thickness, drying, flow, substrate penetration or spatial pigment separation. Those effects belong to a larger painting engine and can remain meaningful even when the same optical mixing primitive is used. Our purpose is to establish a reproducible color component whose assumptions are accessible to that larger system.

2. Related work

Kubelka–Munk theory describes coupled diffuse fluxes in scattering media (Kubelka and Munk 1931; Kubelka 1948). Haase and Meyer applied pigment optics to computer graphics and paint-oriented color selection (Haase and Meyer

1992). Curtis et al. combined layered optical compositing with a separate transport simulation for watercolor (Curtis et al. 1997). These works motivate distinguishing a color mixture from a complete paint simulation.

RGB spectral upsampling chooses a spectrum consistent with limited colorimetric information. Smits uses a compact reflectance basis (Smits 1999). Burns studies smoothness-constrained reconstruction (Burns 2020). Jakob and Hanika represent bounded spectra using a compact nonlinear function space (Jakob and Hanika 2019). These methods address spectral representation, but a recovered reflectance still requires a scattering convention before predicting paint mixtures.

Sochorová and Jamriška introduce concentrations plus additive residuals as a practical RGB-compatible pigment representation, with table-based acceleration and surrogate-pigment treatment of gamut issues (Sochorová and Jamriška 2021). We adopt that published principle with independently specified eight-pigment spectra, a regularized inverse, a tetrahedral encoder and direct reduced-sample spectral decoding. Their proprietary implementation data is not used. Pigmento addresses inference of pigment structure from an image rather than a fixed stateless RGB encoder (Tan et al. 2017).

Artist-inspired RYB interpolation offers intuitive visualization without a spectral material model (Gossett and Chen 2004). Neugebauer-type print models combine area coverages and measured overprints rather than homogeneous pigment concentrations (Hébert and Hersch 2015). We review these alternatives but do not treat them as equivalent paint-volume laws. OKLab (Ottosson 2020) is included as a perceptual baseline. The sRGB conversion definitions follow public color specifications (World Wide Web Consortium, n.d.).

The software review includes Colour, rgb2spec and Spectral.js. Colour is used as an independent color-difference implementation and to obtain attributed standard colorimetric tables (Colour Developers, n.d.). No other mixing library is linked, ported or used as a numerical target. The research log documents access limitations: some historical publications were available only through authoritative abstracts or later primary exposition. The project does not present an exhaustive historical review.

3. Background and problem definition

3.1 From reflectance to display color

Let $R(\lambda)$ be diffuse reflectance, $E(\lambda)$ the illuminant spectral distribution, and $\bar{x}, \bar{y}, \bar{z}$ the CIE 1931 two-degree color-matching functions. Define

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = k \int E(\lambda) R(\lambda) \begin{bmatrix} \bar{x}(\lambda) \\ \bar{y}(\lambda) \\ \bar{z}(\lambda) \end{bmatrix} d\lambda.$$

$$k^{-1} = \int E(\lambda) \bar{y}(\lambda) d\lambda.$$

The normalization makes a perfect diffuser have $Y = 1$. We use D65 daylight and attributed CIE observer data (International Commission on Illumination 2019b, 2019a). A fixed matrix maps XYZ to linear sRGB. Display encoding is applied only after optical calculations and gamut mapping. For encoded channel u , the decoding rule is

$$x = \begin{cases} u/12.92, & u \leq 0.04045, \\ ((u + 0.055)/1.055)^{2.4}, & u > 0.04045. \end{cases}$$

The integral provides three constraints on a wavelength-dependent function. There are generally many metameric reflectances for a given triplet. A reconstruction algorithm therefore selects a prior, even when its colorimetric error is zero. We limit the public API to finite sRGB channels in the unit cube; HDR emission, alpha compositing and arbitrary input profiles are separate concerns.

3.2 Opaque Kubelka–Munk mixtures

Let pigment i have absorption $K_i(\lambda) > 0$, scattering $S_i(\lambda) > 0$ and concentration $c_i \geq 0$, with $\sum_i c_i = 1$. Under the selected concentration mixing assumption,

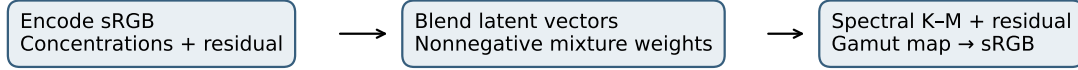
$$K = \sum_i c_i K_i, \quad S = \sum_i c_i S_i, \quad q = K/S.$$

The infinite-thickness reflectance is evaluated in its rationalized form,

$$R_\infty(q) = \frac{1}{1 + q + \sqrt{q(q+2)}}.$$

This is algebraically equivalent to $1 + q - \sqrt{q^2 + 2q}$ but avoids cancellation at large q . Inverting an opaque reflectance gives $q = (1 - R)^2/(2R)$. It does not identify S . In particular, multiplying both K_i and S_i by a pigment-specific factor preserves its isolated color and changes its mixing strength. The scattering prior is thus a consequential modeling choice.

Our implementation assumes an optically thick, homogeneous, diffuse layer. Finite-layer reflectance and transmittance were reviewed but not integrated into the runtime. A finite-layer extension would require thickness, backing reflectance and robust limiting forms, and would not be equivalent to merely changing interpolation weights.



Encoder: $\text{LUT}(\text{sRGB}) \rightarrow \text{concentrations}$; $\text{residual} = \text{linear}(\text{sRGB}) - \text{spectral}(\text{concentrations})$. Reference replaces LUT with direct inversion.

Figure 1: Pipeline. RGB inversion supplies a simplex recipe; its reconstruction residual is carried alongside the recipe during mixing.

4. Method

4.1 Independently constructed pigment basis

We choose eight roles: white, cyan, magenta, yellow, red, green, blue and black. These are synthetic pigment roles, not measurements of specific chemical compositions. Define a logistic spectral transition

$$H_e(\lambda) = [1 + \exp(-(\lambda - e)/w)]^{-1}, \quad w = 22 \text{ nm}.$$

With edges in nanometers, define shapes $B = 1 - H_{510}$, $Y = H_{530}$, $R = H_{600}$, $C = 1 - R$, $G = YC$, and $M = 1 - G$. The basis reflectances are

$$R_i = 0.025 + 0.915b_i, \quad b_i \in (1, C, M, Y, R, G, B, 0).$$

The scattering strengths, in that order, are $(8, 1, 1, 1, 1, 1, 1, 0.5)$. Absorption follows from $K_i = S_i(1 - R_i)^2/(2R_i)$. Positive reflectance bounds avoid singular absorption and guarantee finite optical calculations throughout the simplex. These constants are design priors, not fitted commercial-pigment coefficients. The repository retains the edge sweep that motivated the selected blue/yellow overlap; the remaining initial values are explicitly labeled heuristic.

A larger basis offers more possible recipes and introduces ambiguity. It does not guarantee more realistic behavior. We retain a deterministic recipe prior and assess smaller subsets rather than assuming that the eight-component basis is optimal. No source measurement dataset is silently substituted for the analytic basis.

4.2 Constrained RGB inversion

Let $F(c)$ be the spectral decoder returning unbounded linear RGB, and let L denote conversion from linear RGB to OKLab. For input u and decoded x , we approximately minimize

$$J(c; u) = \|L(F(c)) - L(x)\|_2^2 + \eta \|c - p(u)\|_2^2,$$

subject to the simplex constraints, with $\eta = 0.0002$. The prior p is a piecewise-linear cube decomposition in encoded RGB. White receives the minimum channel and black receives one minus the maximum. The remaining mass is split between one secondary and one primary according to sorted channel differences. For example, when $r \leq g \leq b$, cyan receives $g - r$ and blue receives $b - g$. The other channel orders are symmetric. The prior is continuous, but its derivative changes at channel-order boundaries.

The reference solver uses projected Gauss-Newton steps, analytic derivatives, pivoted dense solves, and backtracking. Differentiation gives

$$\frac{\partial R}{\partial q} = -\frac{R}{\sqrt{q(q+2)}}, \quad \frac{\partial q}{\partial c_i} = \frac{K_i - qS_i}{S}.$$

The OKLab chain rule completes the Jacobian. The normal matrix includes the objective regularizer and a numerical damping term 10^{-5} . We project candidate recipes onto the simplex and accept only objective decreases exceeding 10^{-13} . A projected-gradient fallback is essential: projection of an unconstrained Newton direction need not yield descent on a constrained face. The failure of the initial Newton-only variant was observed in the canonical tests and retained in the decision log.

There are at most 80 outer iterations, 16 Newton backtracking trials, and 24 gradient fallback trials. The fallback starts at step size 8 and halves it. This is a reproducible local algorithm, not a proof of the globally best concentration vector. The weak prior makes the underdetermined problem better specified without eliminating local minima. Independent Python SLSQP calculations are used for exploratory comparisons and ablations; they are not assumed to yield identical recipes.

4.3 Linear-light residual and mixture law

The encoded latent is $z = (c, e)$, where $e = x - F(c)$. For normalized nonnegative weights α_i , the mixture is

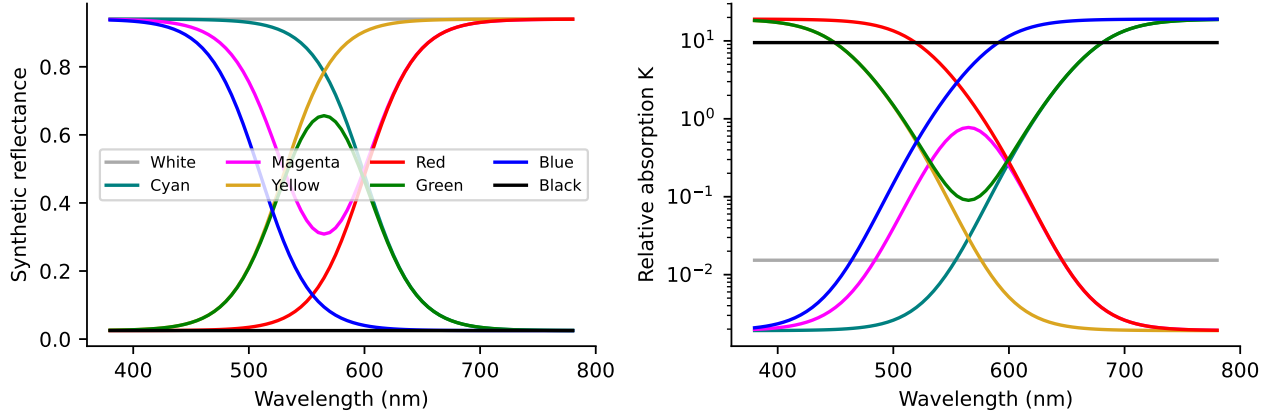


Figure 2: Original synthetic reflectance and derived absorption spectra. White and black are spectrally neutral; their relative scattering differs.

$$\bar{c} = \sum_i \alpha_i c_i, \quad \bar{e} = \sum_i \alpha_i e_i, \quad \bar{x} = F(\bar{c}) + \bar{e}.$$

This construction gives exact reconstruction in real arithmetic because $F(c) + [x - F(c)] = x$. The residual may contain negative components and is not an absorption or scattering coefficient. It supplies a display-space correction where the physical palette is insufficient. All experiments therefore distinguish corrected reconstruction, raw palette approximation and intermediate mixture behavior.

The fast path recomputes the residual after table interpolation using the actual fast decoder. Interpolating residuals stored only at the table nodes would not provide the same identity between nodes. For fixed source latents, concentrations and residuals are affine in the mixture parameter. The nonlinear K–M decoder produces the resulting curved color trajectory.

4.4 Display gamut treatment

The physical decoder and residual can produce out-of-range linear RGB. We retain raw output for diagnostics and map only for display. For a raw vector x , set $y = \text{clamp}(0.2126x_R + 0.7152x_G + 0.0722x_B, 0, 1)$ and $d = x - y\mathbf{1}$. Choose the largest $a \in [0, 1]$ such that $y\mathbf{1} + ad$ lies in the RGB cube. Each positive component of d bounds a by $(1 - y)/d_j$; each negative component bounds it by $-y/d_j$.

This neutral-axis compression is identity inside the gamut and continuous at its boundary. It preserves linear luminance when that luminance is in range. It does not preserve perceptual hue in general. Beyond the available luminance range the output collapses to black or white. We count mapping incidence rather than hiding it as an implementation detail. This simple mapping remains a quality limitation of the current design.

5. Reference and fast implementations

5.1 Reference path

The reference samples 380–780 nm inclusive every 5 nm, using 81 f64 values. Trapezoidal integration weights combine D65, observer functions and the XYZ-to-RGB matrix. A small diagonal normalization maps a perfect diffuser to exactly (1, 1, 1), removing the truncated-tabulation white bias. This is a deliberate approximation to unmodified standard integration. A 1 nm comparison is included to assess wavelength sampling for the smooth spectra used here.

The bundled observer and illuminant subset is obtained through Colour 0.4.6. Observer samples are at 1 nm; the package’s 5 nm D65 samples are linearly interpolated to 1 nm before subsampling. We do not claim byte identity with the current official CIE CSV files. The exact subset checksum, source attribution and transformation are recorded. Standard colorimetric data and derived assets retain their documented CC BY-SA terms, separately from the MIT OR Apache-2.0 source code.

The reference is ground truth only for approximation comparisons within this synthetic model. It is not ground truth for a real paint sample. That distinction applies both to error statistics and to the word reference in the public API.

5.2 Lookup-table encoder

We sample the encoded RGB cube uniformly at resolutions 17^3 , 33^3 and 65^3 . Every node stores eight f32 concentrations generated by the same reference optimizer. No neighbor warm start is used, so generation order cannot affect a node’s initialization. The table is regenerated by a separate executable; no unexplained binary coefficients are required.

At a query, sort the three fractional cell coordinates as $f_a \geq f_b \geq f_c$. The tetrahedron’s four interpolation weights

are $(1 - f_a, f_a - f_b, f_b - f_c, f_c)$. They are nonnegative and sum to one, preserving the recipe simplex. Shared-face agreement gives a continuous encoder even if the optimized node recipes vary sharply. It does not make the encoder differentiable, remove steep local changes, or make an ambiguous inverse unique. A trilinear eight-vertex encoder is included for comparison.

Having eight output pigments does not prevent a 3D encoder table because its domain is RGB. It does prevent a general unrestricted concentration decoder from being represented by an ordinary 3D table. We therefore evaluate the decoder spectrally instead of claiming that all latent dimensions can be indexed by the RGB table.

5.3 Reduced spectral decoder and runtime API

The fast decoder uses 21 samples at 20 nm in f32, with fixed-size stack arrays and precombined colorimetric weights. It evaluates the same positive K-M formula and palette. We measure sampling and floating-point effects separately where possible. No hand-written SIMD, fitted polynomial decoder, neural network, Rayon or GPU kernel is implemented. The compiler’s ordinary optimizations are not labeled a custom vectorized implementation.

The public API accepts checked sRGB colors, pair interpolation and arbitrary nonnegative weighted mixtures. Invalid weights return an error. A convenience pair API clamps t and maps NaN to the first endpoint; a checked alternative rejects invalid t . Pair endpoints are returned exactly. The latent structures keep their internal concentrations private and provide read-only accessors. Raw decode is available for scientific inspection.

A complete RGB mixture decodes three spectra: one for each input’s residual and one for the mixed recipe. Cached latent mixing decodes only the result. Applications should retain a mixer object and cache paint latents where possible. The embedded default table is parsed once and shared immutably. No per-mixture heap allocation, external crate, unsafe code, file operation or network access is required in the runtime.

6. Experimental methodology

The evaluation uses 10000 uniformly sampled encoded-RGB colors and 10000 independent random color pairs with uniformly sampled t . Rust uses an explicitly implemented xorshift64 sequence seeded with 20260929; exploratory Python ablations use NumPy’s documented generator with the same integer seed, which does not imply identical sequences. Sample CSV files accompany the release, allowing metrics to be recalculated without regenerating the random streams.

Reconstruction is assessed in CIEDE2000 (Sharma, Wu, and Dalal 2005), $100\|\Delta\text{OKLab}\|$, and channel error. Colour

0.4.6 provides the independent CIEDE2000 implementation. We report mean, median, 95th percentile, 99th percentile and maximum. We evaluate the uncorrected palette after the same display gamut map rather than measuring only the algebraically restored endpoints. Raw linear channels are also retained.

Mixing comparisons include encoded-sRGB interpolation, linear-RGB interpolation, OKLab interpolation, the fast hybrid, the reference hybrid and the fast decoder without residuals. Twelve canonical pairs cover primaries, secondaries, complements, tints, neutrals and an extreme yellow–blue case. Each trajectory uses 1001 samples. These examples were specified before the final Rust experiment. They are useful diagnostics, not a representative distribution of artists’ paint choices. There is no blinded perceptual study or measured physical target.

The spectral sampling ablation fixes the recorded reference concentrations and reevaluates them at 1, 5, 10, 20 and 40 nm. This isolates the forward approximation from inverse selection. The palette ablation uses 128 common colors and the same independent SLSQP procedure for CMYW, CMYWK, eight pigments, removal of white, and equal white/chromatic scattering. It measures each subset’s uncorrected approximation; residual-corrected reconstruction would hide the effect under investigation.

Runtime tests use one native thread, varying colors and t from a seeded workload, and compiler barriers to prevent dead-code removal. Seven repetitions are recorded. Most modes run 100000 operations per repetition; direct reference inversion is limited to 1000 because of its cost. Timings include the public color-output conversion where that method returns RGB. Lookup-only timings are labeled separately. Table parsing is measured separately from steady-state mixing, and offline generation wall time is recorded. There is no direct benchmarking of Mixbox or other proprietary software.

The host reports AMD EPYC 9V74 80-Core Processor. The build uses Rust 1.75.0, release optimization, one code-generation unit, no LTO and no target-cpu=native flag. The full environment record is included. Host scheduling and frequency are uncontrolled; these are local microbenchmarks, not universal performance guarantees. No browser/WASM throughput is inferred from them.

7. Results

7.1 Endpoint consistency and palette limitations

Corrected reconstruction is near floating-point precision, as expected. The uncorrected palette has substantially larger error. Its worst regions are exposed in the heatmap and top-error CSV rather than excluded as unusual inputs. Saturated display colors are particularly difficult because broad positive-reflectance pigments do not span the display

Table 1: Source reconstruction (CIEDE2000).

Model	Mean	Median	P95	P99	Max
Reference + residual	0	0	0	0	0
Fast + residual	1.19e-07	0	6.86e-07	1.17e-06	2.38e-06
Reference palette only	11.7	9.58	28.4	34.6	42
Fast palette only	11.8	9.65	28.4	35	42.3

cube. This model deliberately trades unrestricted RGB gamut for an explicit material prior.

The forward implementation cross-check, evaluating the recorded Rust recipes in independent Python f64 code, has maximum linear-channel difference 1.31e-13. This validates agreement of equations, coefficients and integration for identical recipes. It does not validate the recipes against physical measurements. The 20 nm f32 check, including concentration rounding, has maximum linear-channel discrepancy 3.2e-07 against f64. The wavelength study further separates arithmetic accuracy from modeling accuracy.

7.2 Canonical mixtures and gamut mapping

The selected yellow–blue trial bends into green; red–blue gives a dark purple rather than an additive average. Complementary examples reduce chroma, and adding white changes both the amount and trajectory of lightening because scattering is mixed separately from absorption. The cyan–magenta midpoint is muted rather than a clean saturated blue, and the red–white tint shifts markedly toward peach. These are visible compromises of this chosen basis and residual convention. Nevertheless, the examples do not establish material fidelity. A paint identified only as red or blue could have a different spectral tail and very different mixing behavior.

8.42% of the random fast mixtures require display gamut mapping. The raw linear output range across those trials is $[-0.209, 1.061]$. This is evidence that residual correction and the physical basis sometimes disagree substantially. The mapper maintains bounded output and continuity, but may compress chroma and distort hue. Those effects appear in the trajectory plots and must be addressed before claiming a high-fidelity professional paint simulation.

7.3 Table approximation and spectral sampling

These errors compare full displayed mixtures against the direct reference model, rather than comparing only table-node reconstructions. The default 33^3 table is a memory/approximation compromise. Each node occupies 32 bytes, giving 157,216, 1,149,984 and 8,788,000 bytes of concentration payload at the three resolutions; the binary header adds eight bytes. The maximum 33^3 discrepancy of 40.1 CIEDE2000 is a material failure of uniform approximation. Low average error does not make the table a

reliable substitute in every RGB region. The worst examples involve large changes in selected recipe and residual, exposed in the supplementary worst-case figure. Increasing the table resolution changes both local interpolation and how closely the fast path follows the reference inverse’s recipe choices. A local inverse is not necessarily a globally smooth target, so monotonic quality improvements should not be assumed.

The 20 nm forward model, evaluated in f64 at fixed recipes, has mean CIEDE2000 0.0826 and 95th-percentile 0.149 relative to 5 nm. This isolates sampling error from the LUT’s inverse approximation. Since the analytic spectra are smooth and broad, the coarse sampling is more successful than it would necessarily be for narrow-band or measured interference pigments. The 1 nm comparison is a numerical check of this restricted basis, not a claim that all paint spectra can be represented at 20 nm.

7.4 Smoothness

All recorded canonical displayed samples are finite. The largest adjacent-step difference among fast canonical trajectories is 7.75 Δ EOK100 at the tested spacing. A separate 200-pair, 1,001-step random test has largest adjacent step 6.42. The random maximum is perceptually substantial over a small parameter change and fails a stronger requirement of uniformly gentle visual transitions. These sampled bounds do not prove perceptual smoothness for all inputs. The mathematical argument is more specific: convex concentration interpolation stays in a positive optical domain; K–M decoding and the residual are continuous; the gamut map is continuous. Derivative kinks can still arise from gamut boundaries, table faces or channel-order changes.

The tests separately probe source reconstruction, exact pair endpoints, symmetry, nonnegative normalized weights, nonfinite input handling, malformed tables, and crossings of LUT cell boundaries. Eight-bit output can still band even when the floating-point curve is continuous. Retaining floating-point canvas data and applying an appropriate display/dithering strategy remains the host application’s responsibility.

7.5 Runtime performance

The median full fast RGB operation costs 602 ns and the cached latent operation costs 226 ns on this host. The cost difference follows the number of spectral evaluations and input transfer functions, not a change in the mixture

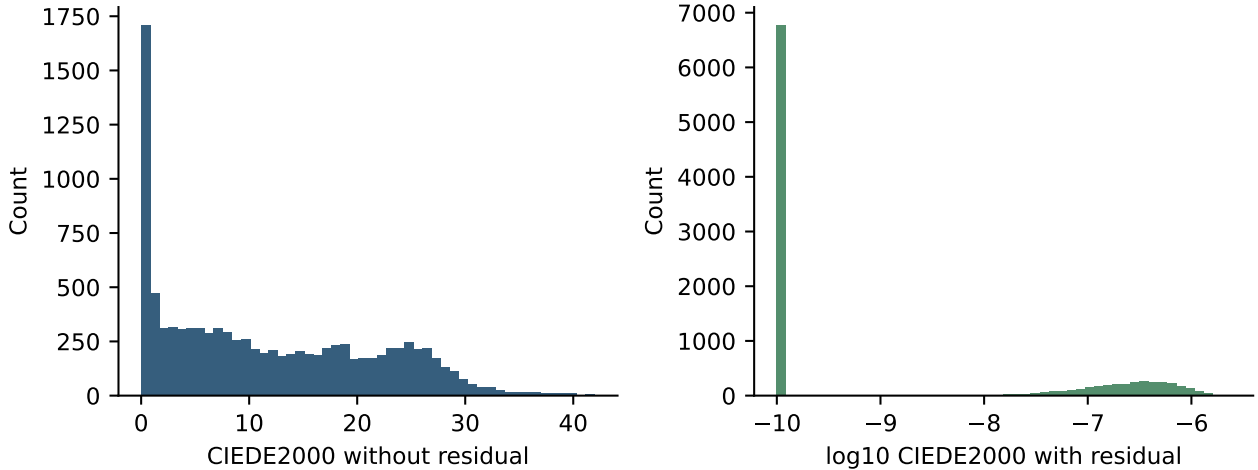


Figure 3: Reconstruction error distributions. The left plot measures the palette alone; the right shows the tiny numerical error after residual correction on a logarithmic scale.

Table 2: Canonical fast-model midpoint colors.

Pair	Fast midpoint (sRGB8)
black + white	(141, 141, 141)
blue + orange	(103, 87, 91)
blue + white	(107, 188, 255)
cyan + magenta	(148, 101, 130)
extreme + yellow + blue	(60, 174, 124)
magenta + yellow	(237, 120, 116)
red + blue	(88, 0, 71)
red + green	(137, 100, 39)
red + white	(255, 166, 124)
red + yellow	(242, 118, 14)
yellow + blue	(49, 144, 112)
yellow + purple	(162, 130, 134)

law. Lookup alone is faster but is not the complete mixing operation and must not be advertised as such. The default table parsing median is 0.267 ms. Offline generation durations are retained in the performance dataset; they are excluded from steady-state timing.

8. Ablations and negative findings

The subset experiment measures representation power under a common optimizer and source sample set. Removing components can force a larger residual, but a larger basis can also create additional recipe choices. We therefore interpret improvements in raw reconstruction as increased colorimetric flexibility, not automatic improvement in mixture realism. Removing white is particularly relevant to tints, while equalizing scattering changes mixing strength without necessarily changing the isolated pigment colors.

The first full-spectrum trial used a bounded, first-difference-regularized reflectance reconstruction, followed by constant-scattering K–M. Its black–white midpoint was approximately (7,7,7) in 8-bit sRGB. The reconstructed black’s

large K/S dominated the mixture. This is a concrete example of why colorimetric reconstruction alone is insufficient. The selected palette represents black with finite absorption, an explicit relative scattering, and a residual for the remaining endpoint mismatch.

The initial palette put the blue edge at 500 nm and the yellow edge at 490 nm. Its corrected yellow–blue midpoint was (19,128,145), too cyan for the target example. Moving both edges to 530 nm improved green overlap but caused the red–blue trial to collapse to black after residual correction and mapping. We retained a 510 nm blue edge and 530 nm yellow edge as a compromise. This is limited tuning on diagnostic examples and is openly identified as such; the subsequent random trials assess behavior outside those examples.

The first Rust inversion used only projected unconstrained Gauss–Newton steps. A canonical hue test failed because projection could destroy descent on an active constraint face. Adding projected-gradient fallback corrected that numerical failure. We did not weaken the test to accept the faulty result. The independent Python optimizer helped

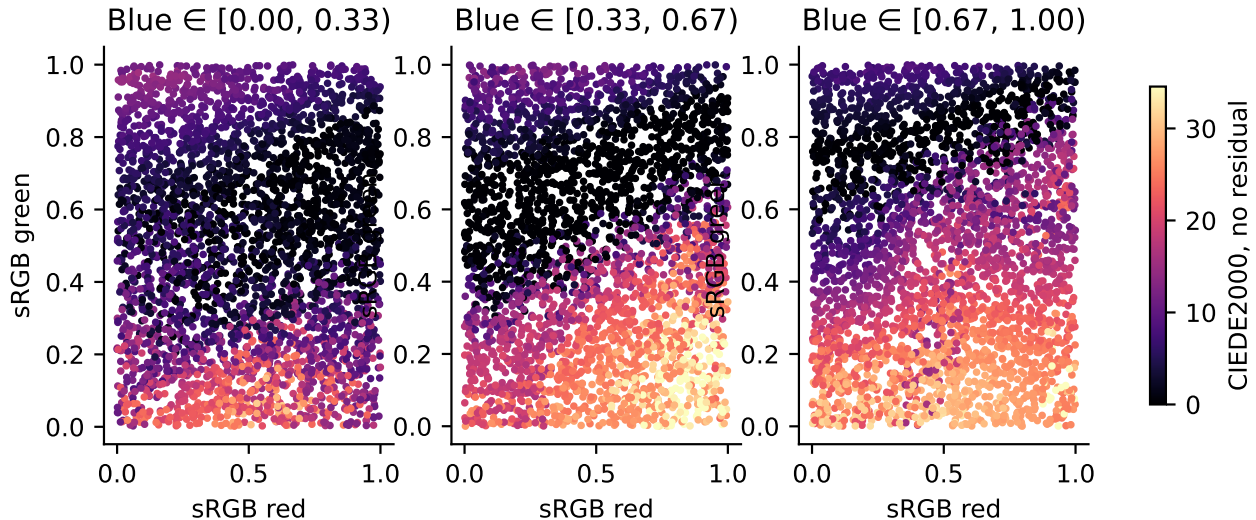


Figure 4: Systematic view of uncorrected palette error over random RGB samples, divided into blue-channel slices. No sample is removed for having high error.

Table 3: Mixture error relative to direct reference (CIEDE2000).

Encoder	Mean	Median	P95	P99	Max
n17	0.706	0.184	2.55	10.9	43.8
n33	0.409	0.0918	1.19	6.48	40.1
n65	0.265	0.0631	0.774	4.19	36.7
tri33	0.438	0.0994	1.38	6.93	40.5

identify that the problem was inversion, rather than the K–M forward equations.

No negative conclusions are claimed for neural encoders, polynomial decoders, SIMD, GPU implementations or measured pigment fitting: these were reviewed or proposed, but not experimentally implemented. Distinguishing a rejected hypothesis from an untested option is necessary for a reproducible research record.

9. Discussion

The most favorable reconstruction statistic is also the easiest to misinterpret. A residual makes arbitrary RGB reconstruction nearly exact by construction. This is useful engineering: pasted artwork and arbitrary color-picker values can be accepted without an endpoint color shift. It does not show that a small physical palette spans RGB, that the inferred spectrum is correct, or that intermediate colors match real mixtures. Reporting the palette-only error and mapping incidence is therefore essential.

The eight-component palette retains more flexible recipes than a four-component palette, but RGB provides fewer constraints than the recipe has degrees of freedom. Regularization chooses a convention. The LUT then approximates that convention. There is no reason to expect two independent solvers with different priors to produce the same

intermediate color even if both reconstruct the endpoints. This is a design space for artistic evaluation, not an error that can be removed simply by increasing numerical precision.

Spectral work remains inexpensive enough to be useful after the inverse is precomputed. The current CPU implementation spends a fixed amount of work on a small number of smooth bands. A polynomial or neural decoder could reduce that work further, but would need to preserve positivity, endpoint correction, and the observed curve quality. The measured cached-latent path establishes a concrete baseline against which such complexity should be justified.

The representation also clarifies a canvas-design decision. An RGB-only canvas repeatedly discards the inferred recipe after displaying each result. Re-encoding can produce a new recipe. Consequently a sequence of RGB pair operations is not generally associative, even though normalized linear combinations of retained latents are. An application that needs persistent material history should store latent recipes and paint amounts, or spectral state, alongside its display buffer. A drop-in RGB mixer cannot preserve information the host discards.

Adding white is especially revealing. Raising the white scattering coefficient produces stronger tinting without changing the isolated white reflectance. This behavior is consis-

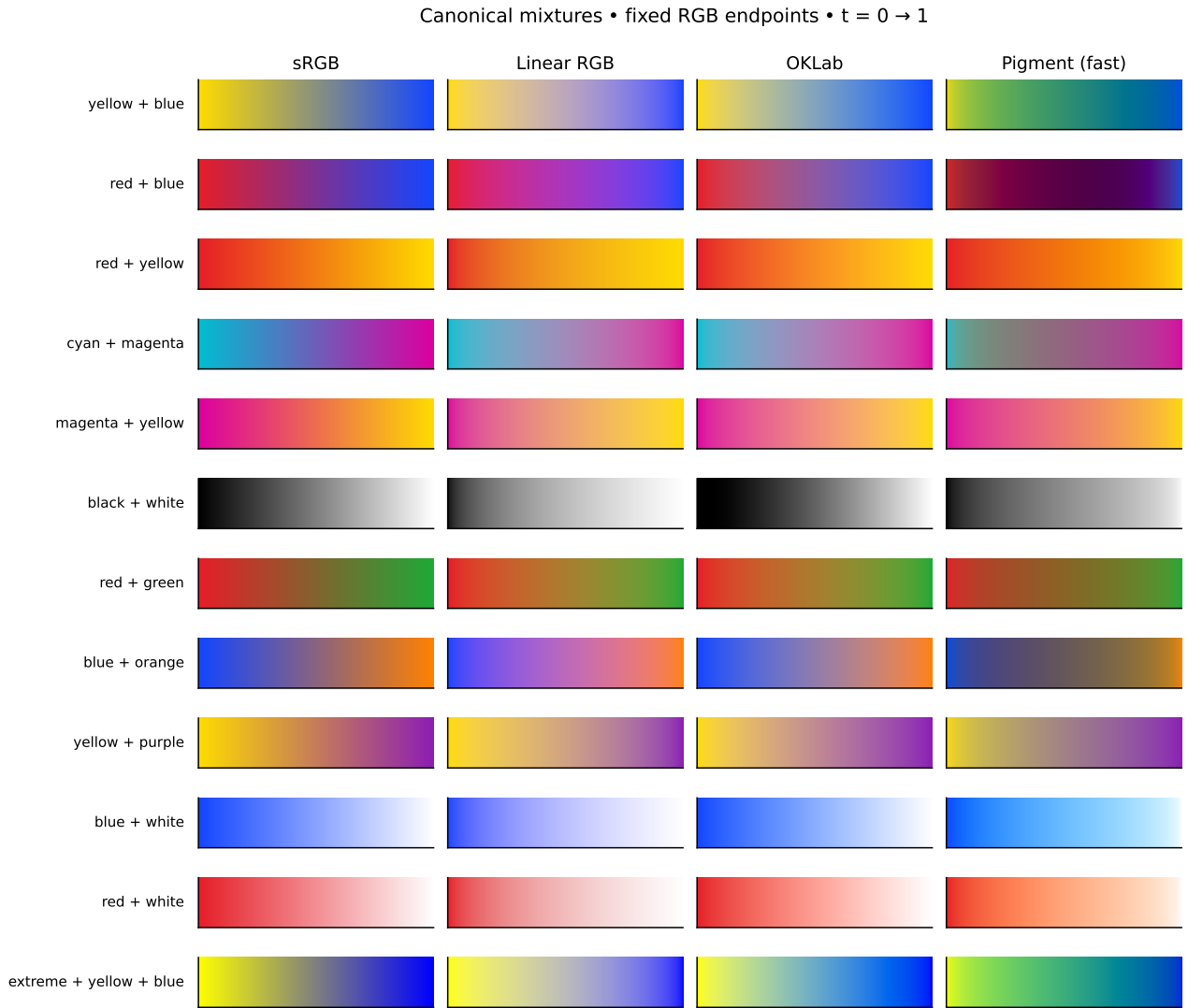


Figure 5: Canonical gradients, with identical endpoints and mixture parameters across methods. All rows, including difficult complement and tint cases, are shown.

tent with the model’s separation of absorption and scattering, but the selected coefficient is not calibrated to titanium white. A future dataset should contain either separate optical coefficients or enough thickness/backing/mixture observations to estimate them. A collection of opaque RGB swatches is not sufficient for this calibration.

10. Limitations

The principal limitation is the lack of measured pigment-mixture validation and artist evaluation. The spectra, scattering strengths and selected edge positions are synthetic priors. Some display colors require large residuals; gamut mapping is frequent enough to matter and can alter perceptual hue. The local inverse has no global-optimality guarantee, and LUT continuity does not guarantee gentle gradients with respect to every possible input perturbation.

The model assumes a homogeneous, opaque, diffuse layer under one illuminant and observer. It does not represent finite paint thickness, translucent glazing, gloss, fluorescence, metallic or interference pigments, particle-size-dependent scattering, granulation, or spatial pigment segregation. It does not predict wetness, drying, canvas absorption or brush transport. The RGB residual is not a spectrum, so relighting the corrected color under a different illuminant is not physically defined by the current representation.

Numerical evaluation is restricted to the declared synthetic basis and sample distributions. Random RGB samples are not a distribution of real-world paints. The fine-sampling comparison does not cover sharply structured measured spectra. Performance is native, scalar-source, single-host evidence; the WASM CI target is a portability check, not a browser benchmark. The release is not presented as externally reviewed, deployed in a production painting program or peer-reviewed.

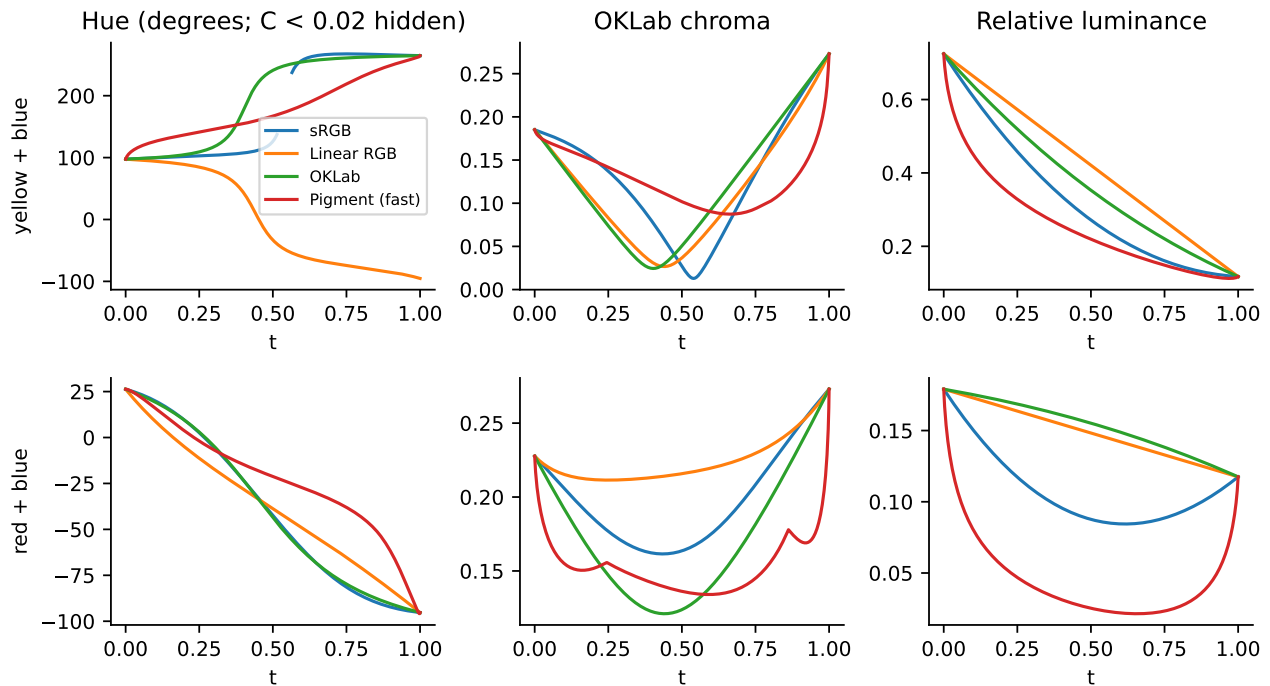


Figure 6: Hue, chroma and luminance trajectories. Hue is hidden below OKLab chroma 0.02 because its angle is unstable near neutral. Curvature and nonmonotonic luminance can be legitimate outcomes, but are not independently validated paint behavior.

Table 4: Single-thread native timing; lookup is not a full mix.

Operation	Median ns	Million/s
sRGB	19.2	52
linear RGB	160	6.27
OKLab	245	4.07
reference RGB	2.39e+06	0.000418
fast RGB	602	1.66
fast cached latent	226	4.42
tetrahedral lookup	34.8	28.7
reference cached latent	835	1.2
trilinear lookup	34	29.4

11. Future work

The highest-value next experiment is a permissively redistributable dataset of measured mixtures, including tints and multiple application thicknesses or backings. This would support fitting K and S separately, comparing spectral errors as well as colorimetric errors, and testing whether the synthetic palette’s successful examples survive material-specific calibration. A blinded artist study could then evaluate useful behavior beyond named-color expectations.

The inverse could incorporate neighborhood regularization across the LUT, a constrained surrogate optimizer, or a learned predictor with a reproducible training set. Such methods should be assessed on recipe stability and intermediate mixtures, not just endpoint RGB. A larger palette should be justified by measured gains and manageable am-

biguity. Alternative gamut mapping should be evaluated on the currently recorded out-of-range cases rather than only visually favorable gradients.

For performance, batching and SIMD are straightforward candidates because wavelengths and independent pixels admit parallel work. GPU and WASM implementations require their own precision and timing studies. A fitted decoder may reduce runtime spectral integration, but should retain a reference spectral mode for validation. Finite-layer K – M , backing interactions and a physically specified binder would be required for glazing. Wet-on-wet and oil- or watercolor-specific behavior would additionally require transport and material-state models.

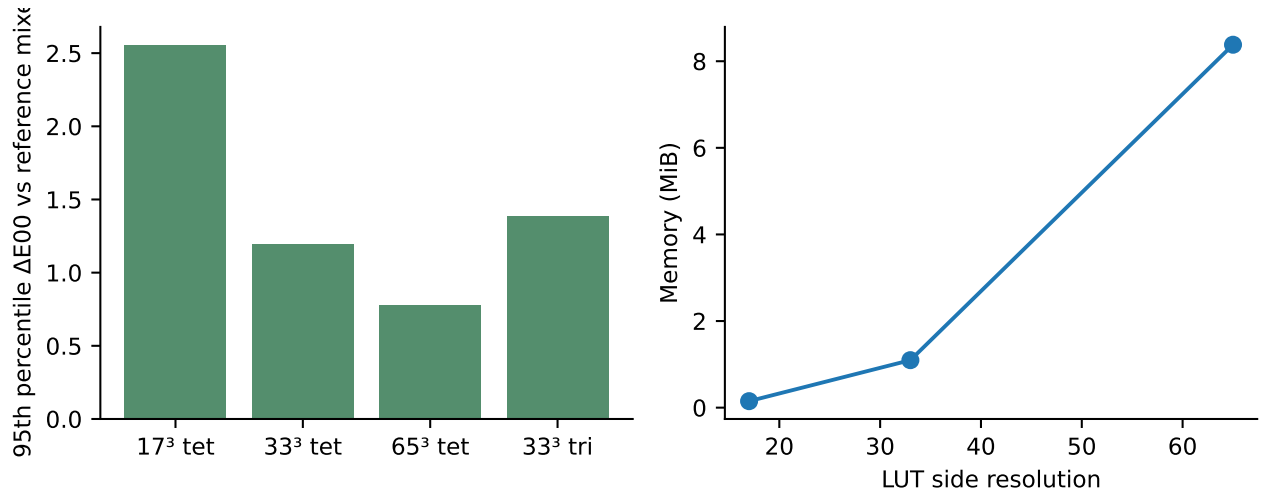


Figure 7: LUT approximation versus reference mixtures, and table memory. The trilinear variant uses the same 33^3 node data.

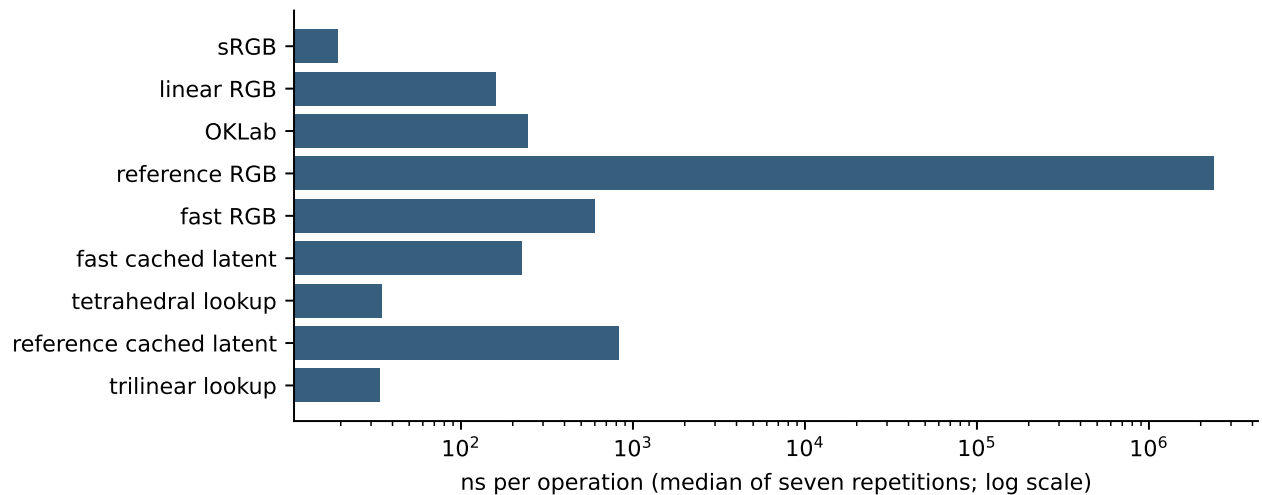


Figure 8: Single-thread native operation costs, with logarithmic horizontal scale. Direct inversion, full RGB mixing, cached latent mixing, and lookup alone are different workloads.

12. Conclusion

This study implements an independent, reproducible RGB pigment-mixing research library using public K–M equations and a published concentration-plus-residual representation. The selected synthetic basis produces the intended green and purple canonical mixtures and supports arbitrary sRGB endpoints with near-floating-point reconstruction error. That endpoint result is algebraic, while the uncorrected palette error of 11.7 mean CIEDE2000 exposes the basis’s limitations.

The default fast model reaches 1.66 million complete RGB mixtures per second, and 4.42 million cached-latent mixtures per second on the measured host. Its random-mixture mean discrepancy from the reference is 0.409 CIEDE2000. The project therefore establishes a usable implementation and performance baseline, together with clear unresolved

quality risks: synthetic material priors, large residuals for some inputs, inverse ambiguity and gamut mapping. It is suitable for integration experiments and further validation. A claim of calibrated or professional-grade paint fidelity would require evidence this study does not yet provide.

Reproducibility and authorship note

`python3 tools/reproduce.py` regenerates model arrays, all three encoder tables, tests, raw experiment outputs, metrics, figures and this manuscript’s result tables. The bundled data permit offline runtime builds. Python dependencies and model settings are pinned or recorded; `results/environment.json` and the source/data manifest identify the reported configuration. The accompanying Markdown, TeX, bibliography and CSV files are the

Table 5: Palette and scattering ablations, without residual (CIEDE2000).

Basis / S variant	Mean	Median	P95	Max
CMYW	23.6	23.2	49.6	64.3
CMYWK	15.7	15.1	28.6	30.2
eight	11.7	7.06	29.7	38.1
no white	12.1	7.68	30	38.1
equal scattering	11.7	7.06	29.7	38.1

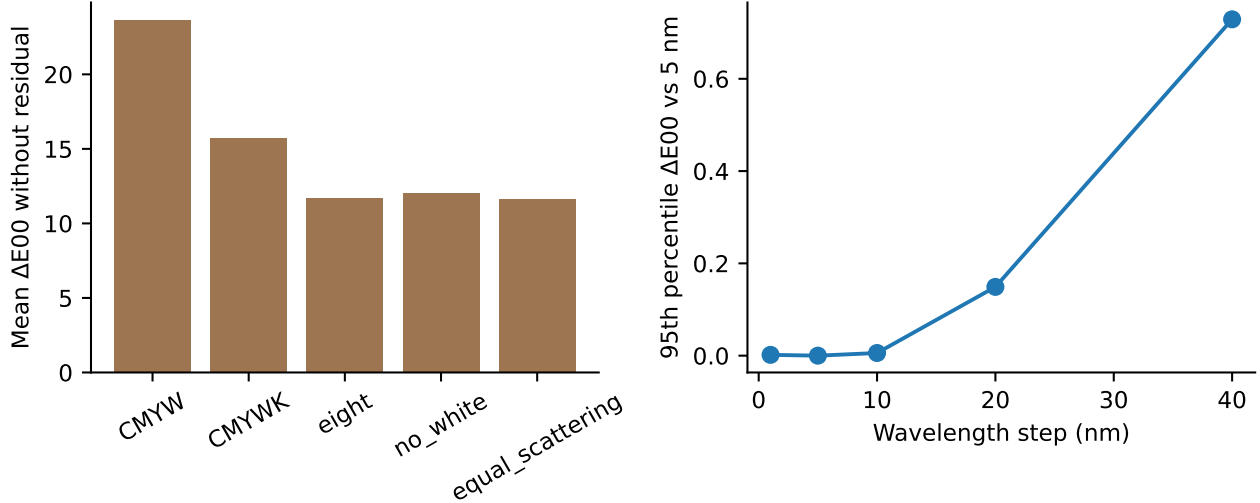


Figure 9: Palette/scattering ablations and fixed-recipe spectral-sampling error. The left experiment uses the independent SLSQP solver on a common 128-color set.

primary research record. This is an AI-assisted implementation and manuscript prepared for the commissioning user, with measurements produced by executable repository experiments. It has not undergone external peer review.

References

- Burns, Scott Allen. 2020. “Numerical Methods for Smoothest Reflectance Reconstruction.” *Color Research and Application* 45: 8–21. <https://doi.org/10.1002/col.22437>.
- Colour Developers. n.d. “Colour: Colour Science for Python, Version 0.4.6.” <https://github.com/colour-science/colour>.
- Curtis, Cassidy J., Sean E. Anderson, Joshua E. Seims, Kurt W. Fleischer, and David H. Salesin. 1997. “Computer-Generated Watercolor.” In *Proceedings of SIGGRAPH*, 421–30. <https://grail.cs.washington.edu/projects/watercolor/>.
- Gossett, Nathan, and Baoquan Chen. 2004. “Paint Inspired Color Mixing and Compositing for Visualization.” In *IEEE Symposium on Information Visualization*, 113–18. <https://doi.org/10.1109/INFOVIS.2004.52>.
- Haase, Chet S., and Gary W. Meyer. 1992. “Modeling Pigmented Materials for Realistic Image Synthesis.” *ACM Transactions on Graphics* 11 (4): 305–35. <https://doi.org/10.1145/146443.146452>.
- Hébert, Mathieu, and Roger D. Hersch. 2015. “Review of Spectral Reflectance Models for Halftone Prints: Principles, Calibration, and Prediction Accuracy.” *Color Research and Application*. <https://doi.org/10.1002/col.21907>.
- International Commission on Illumination. 2019a. “CIE Standard Illuminant D65.” <https://doi.org/10.25039/CIE.DS.hjfjmt59>.
- . 2019b. “Colour-Matching Functions of CIE 1931 Standard Colorimetric Observer.” <https://doi.org/10.25039/CIE.DS.xvudnb9b>.
- Jakob, Wenzel, and Johannes Hanika. 2019. “A Low-Dimensional Function Space for Efficient Spectral Upsampling.” *Computer Graphics Forum* 38 (2). <https://doi.org/10.1111/cgf.13626>.
- Kubelka, Paul. 1948. “New Contributions to the Optics of Intensely Light-Scattering Materials. Part i.” *Journal of the Optical Society of America* 38 (5): 448–57. <https://doi.org/10.1364/JOSA.38.000448>.
- Kubelka, Paul, and Franz Munk. 1931. “Ein Beitrag Zur Optik Der Farbanstriche.” *Zeitschrift für Technische Physik* 12: 593–601.
- Ottosson, Björn. 2020. “A Perceptual Color Space for Image Processing.” <https://bottosson.github.io/posts/oklab/>.
- Sharma, Gaurav, Wencheng Wu, and Edul N. Dalal. 2005. “The CIEDE2000 Color-Difference Formula: Implemen-

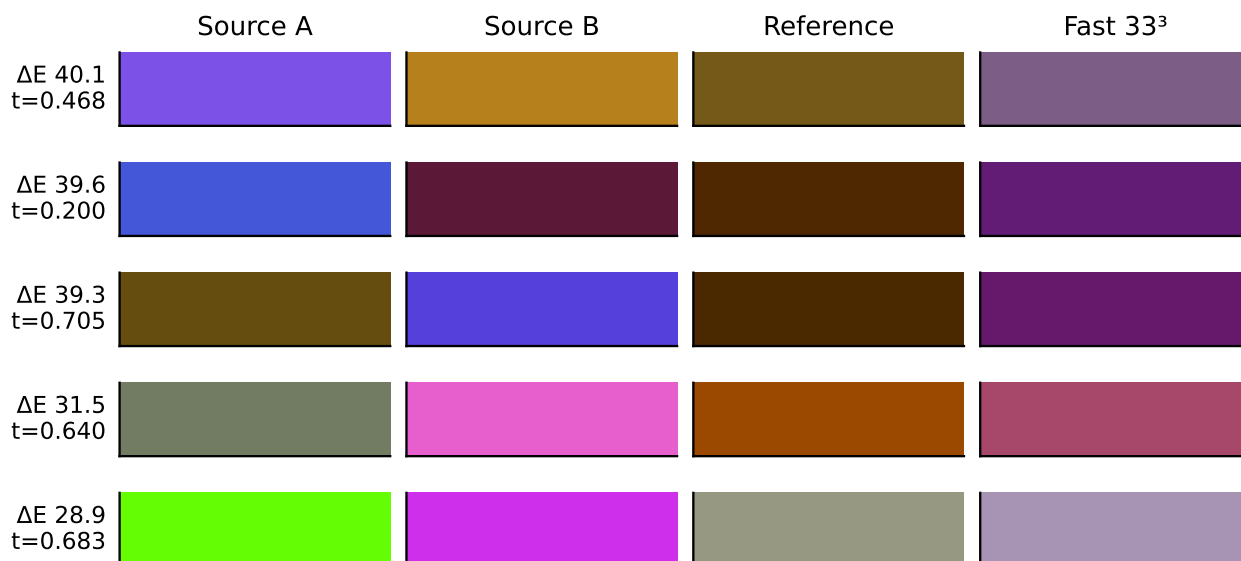


Figure 10: The five largest default-LUT discrepancies in the random-pair experiment. Both outputs are model predictions; neither is a measured-paint target. The large disagreements expose a weakness that mean error alone hides.

- tation Notes, Supplementary Test Data, and Mathematical Observations.” *Color Research and Application* 30 (1): 21–30. <https://doi.org/10.1002/col.20070>.
- Smits, Brian. 1999. “An RGB-to-Spectrum Conversion for Reflectances.” *Journal of Graphics Tools* 4 (4): 11–22. <https://doi.org/10.1080/10867651.1999.10487511>.
- Sochorová, Šárka, and Ondřej Jamriška. 2021. “Practical Pigment Mixing for Digital Painting.” *ACM Transactions on Graphics* 40 (6): 234:1–11. <https://doi.org/10.1145/3478513.3480549>.
- Tan, Jianchao, Stephen DiVerdi, Jingwan Lu, and Yotam Gingold. 2017. “Pigmento: Pigment-Based Image Analysis and Editing.” <https://arxiv.org/abs/1707.08323>.
- World Wide Web Consortium. n.d. “CSS Color Module Level 4.” <https://www.w3.org/TR/css-color-4/>.